

Alex McIntosh

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SUMMARY

Operationalizing **interpretable machine learning, explainable AI, and observable LLM networks** reliably at scale.

EXPERIENCE

Pioneer Management Consulting , Consultant, Artificial Intelligence Designing custom AI architectures and developing conversational analytics to drive sustainable enterprise AI adoption.	Minneapolis, MN June 2026 – present 2 months
- Design AI architectures delivering scalable predictive analytics and anomaly detection for enterprise clients. - Develop AI-powered conversational analytics and generative domain bots. - Implement model explainability into technical designs to ensure client trust and mitigate risk.	
Optum , AI/ML Engineer Associate Architected and built enterprise-scale agentic voice AI systems with end-to-end observability, delivering AI tools to 400K+ employees.	Eden Prairie, MN Aug 2025 – June 2026 11 months
- Shipped an agentic voice AI frontend using LiveKit to expose a conversational frontend to a multi-agent system - Automated performance evaluation for LLM agents and RAG pipelines - Shipped a multi-agent system using A2A, MCP, and ADK to automate admin tasks	
Optum , Data Engineer Associate Architected and deployed enterprise-scale agentic AI systems with end-to-end observability, enhancing productivity for over 400K employees	Eden Prairie, MN Aug 2024 – Aug 2025 1 year 1 month
- Automated LLM evaluation with scoring pipelines to monitor consistency, accuracy, safety, latency, and fairness - Refactored Fabric Spark ETL into reusable patterns, cutting failure rates from 12-15% to <1% - Mentored an intern team that won first place in the Optum-wide Shark Tank competition with an agentic AI solution - Instrumented agents with OpenTelemetry; added tracing and metrics for prompt response, agent calling, and tool use	
Optum , Data Science and Engineering Associate Developed and operationalized ML models and multi-agent systems to drive analytics and improve enterprise decision-making.	Eden Prairie, MN July 2023 – July 2024 1 year 1 month
- Trained a state-of-the-art time-series forecasting model (Prophet + Random Forest) to detect operational anomalies with >90% precision. - Orchestrated multi-agent AI systems to perform statistical analysis dashboard generation - Delivered a conversational AI self-service analytics solution - Enabled observability into millions of updating claims records with Kafka and MLFlow - Evaluated ML models for accuracy, bias, and operational readiness	
Constelleum LLC , Software Engineer Designed and developed a physically accessible multiplayer online video game	Mar 2020 – July 2023 3 years 5 months
- Engineered adaptive control schemes enabling inclusive gameplay - Automated client/server test pipelines - Analyzed event log data to inform balance and engagement metrics	
Stevens County DAC , Employment Specialist	Morris, MN Jan 2022 – May 2022

Led vocational training and program management for individuals with disabilities - Directed programs across 8 job coaches to execute tailored employment programs	5 months
Stevens County DAC , Job Coach	Morris, MN
Coached over 30 individuals in employment skills and workplace readiness	Oct 2021 – Jan 2022
• Developed personalized enrichment activities • Partnered with local employers to improve retention rates	4 months
Prairie Community Services , Program Coordinator	Morris, MN
Led a staff of 12 delivering residential services to individuals with disabilities	Feb 2019 – Feb 2020
• Directed training and compliance for service staff • Managed service plan implementation across 3 residential homes	1 year 1 month
Prairie Community Services , Direct Support Professional	Morris, MN
Provided direct care, daily living assistance, and community integration support.	Feb 2018 – Feb 2019
	1 year 1 month

EDUCATION

Mas-ter of Science	Georgia Institute of Technology , Computer Science	July 2027
	• Machine Learning specialization • Knowledge Based Artificial Intelligence • Deep Learning • Reinforcement Learning	
Micro-Mas-ters	MITx , Statistics and Data Science	Dec 2023
	• Data Analysis • Machine Learning • Statistics and Probability	
Bach-elor of Arts	Saint John's University , Physics	May 2021
	• Minored in Music and Mathematics • Phi Beta Kappa Honor Society Member • Men's Chorus President • Quantum Mechanics • Thermodynamics • Statistical Mechanics • Special Relativity • Mathematical Modeling • Linear Algebra	

CERTIFICATIONS

Microsoft Certified: Fabric Data Engineer Associate	Microsoft Fri Jun
Microsoft Certified: Azure Fundamentals	Microsoft Fri Aug
Microsoft Certified: Azure Data Scientist Associate	Microsoft Wed Jan
Data Science Professional Certificate	HarvardX Fri Dec

Applied Data Science with Python Specialization

University of Michigan
Mon Nov

Mathematics for Machine Learning

Imperial College London
Fri Jan

SQL for Data Science

University of California, Davis
Sun Feb

CS50P: Introduction to Programming with Python

Harvard University
Tue Apr

CS50AI: Introduction to Artificial Intelligence with Python

Harvard University
Wed Aug

CS50SQL: Introduction to Databases with SQL

Harvard University
Wed Aug

PROJECTS

Personal Portfolio

- Configuration-driven portfolio website and PDF resume generator.
- Automated build pipeline with GitHub Actions for multi-format export.

SimLM Review - CS7643 Final Project

- Simulation-guided large language model for enhanced context engineering

ARC AGI

- Computer-vision enhanced agent-based solver for the ARC AGI challenge, a benchmark for artificial general intelligence

LANGUAGES FRAMEWORKS

Languages & Frameworks: Python, PySpark, SQL, R

MACHINE LEARNING AI

Machine Learning & AI: PyTorch, TensorFlow, Statistics and Probability, Data Science and Predictive Analytics, Natural Language Processing (NLP), Simulation, Google Vertex AI, MLFlow

LLM AGENTS

LLM & Agents: Model Context Protocol (MCP), Agent Development Kit (ADK), Agent-to-Agent (A2A) Protocol, Semantic Kernel, Retrieval-Augmented Generation (RAG)

DATA ENGINEERING ANALYTICS

Data Engineering & Analytics: Data Visualization, Data Engineering, Microsoft Fabric, Data Bricks, Delta Lake, AirFlow, ChromaDB

CLOUD INFRASTRUCTURE

Cloud & Infrastructure: Git, Docker, Microsoft Azure, Google Cloud Platform

MLOPS OBSERVABILITY

MLOps & Observability: OpenTelemetry